

Solution Requirements

Date	04 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users register and log in securely using JWT-based authentication with hashed passwords (bcrypt).	High
2	Authorization levels	Each user can only access their own financial profile, debts, expenses and reports (route-level protection).	High
3	External interfaces	Google Gemini API for AI analysis and chat; ReportLab for PDF generation; Axios-based REST calls between React frontend and FastAPI backend.	High
4	Transactions processing	Processes user input to create/update/delete debts and expenses, recompute dashboard metrics, and generate AI insights.	High
5	Reporting	Displays dashboard analytics, Financial Health Score, spending breakdowns, and generates downloadable PDF financial reports.	Medium
6	Business rules, etc.	Financial Health Score (0-100) is computed using Debt-to-Income Ratio, Expense Ratio and Savings Rate; repayment plans follow Snowball or Avalanche logic.	High
7	Compliance to laws or regulations	Stores only necessary financial data per user account, hashes passwords, and isolates data access per authenticated user.	Medium
8	Other	Offline AI demo mode returns placeholder responses when GEMINI_API_KEY is not configured, keeping all features demonstrable.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Dashboard and AI responses should load quickly for a smooth user experience.	Response within 3 seconds
2	Scalability	Backend should support multiple concurrent users and be portable from SQLite to PostgreSQL.	Configurable via SQLAlchemy without code change
3	Security & Data Privacy	Passwords hashed with bcrypt; JWT tokens for session security; per-user data isolation.	HTTPS APIs, hashed credentials
4	Reliability & Availability	Application should work consistently, including offline AI demo mode when no API key is set.	99% availability during usage
5	Usability & Accessibility	Simple, chart-driven React interface for tracking debts, expenses and AI guidance.	User completes core tasks in a few clicks
6	Other	Maintainable modular backend (routers, services) and component-based React frontend.	Easy to extend with new routers/pages